ENGR 421/DASC 521: Introduction to Machine Learning
Homework 01: Naive Bayes Classifier
Deadline: November 2, 2026, 11:59 PM

In this homework, you will implement a naive Bayes classifier using Python. Here are the steps you need to follow:

1. Read Section 5.7 from the textbook.

2. You are given a multivariate classification data set, which contains 310 and 100 bird sound recordings. Each recording is represented as a 20-dimensional real vector after extracting mel-frequency cepstral coefficients (MFCCs). These recordings are from 3 distinct bird types, namely, 1: `Columba palumbus`, 2: `Parus major`, and 3: `Turdus merula`. You are provided with four data files:

   a. `bird_sounds_features_train.csv`: training recordings,

   b. `bird_sounds_labels_train.csv`: training labels,

   c. `bird_sounds_features_test.csv`: test recordings,

   d. `bird_sounds_labels_test.csv`: test labels.

3. Calculate the prior probability estimates $\widehat{\Pr}(y = 1)$, $\widehat{\Pr}(y = 2)$, and $\widehat{\Pr}(y = 3)$ using the data points in the training set. (20 points)

```
class_priors = estimate_prior_probabilities(y_train)
print(class_priors)
```

```
[0.45483871 0.2516129  0.29354839]
```

Hint: You can use the following equation to calculate the prior probability estimates.

$$\widehat{\Pr}(y = c) = \frac{\sum_{i=1}^{N} 1(y_i = c)}{N} = \frac{N_c}{N}$$

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4. Calculate the model parameter estimates $\widehat{\mu}_{1,1}, \widehat{\mu}_{1,2}, \ldots, \widehat{\mu}_{1,20}$, $\widehat{\sigma}_{1,1}, \widehat{\sigma}_{1,2}, \ldots, \widehat{\sigma}_{1,20}$, $\widehat{\mu}_{2,1}, \widehat{\mu}_{2,2}, \ldots, \widehat{\mu}_{2,20}$, $\widehat{\sigma}_{2,1}, \widehat{\sigma}_{2,2}, \ldots, \widehat{\sigma}_{2,20}$, $\widehat{\mu}_{3,1}, \widehat{\mu}_{3,2}, \ldots, \widehat{\mu}_{3,20}$, and $\widehat{\sigma}_{3,1}, \widehat{\sigma}_{3,2}, \ldots, \widehat{\sigma}_{3,20}$ using the data points in the training set. (20 points)

```
means, deviations = estimate_means_and_deviations(X_train, y_train)
print(means)
print(deviations)
```

```
[[-3.81740014e+02  9.94915213e+01  4.44353567e+00  2.30017752e+01
  -1.63074856e+01 -1.19898758e+00 -1.20483036e+01 -1.09408851e+01
  -1.13572162e+01 -5.44665585e+00 -5.85348980e+00  1.67261042e+00
   3.02231022e+00  6.75906065e+00  4.95381969e+00  5.47144031e+00
   2.58078126e+00  3.05536417e+00 -6.79631894e-02  1.28286533e+00]
```

```
 [-4.72688192e+02  6.96392755e+00 -2.23979249e+01  2.81474312e+01
  -2.00854556e+01  5.27805291e+00 -1.10059776e+00 -4.46584731e+00
  -5.11166996e+00  1.35354378e+00 -7.08731232e+00  2.33448538e-01
  -1.88297413e+00  1.04188268e+00  7.24046597e-01 -7.08887987e-01
   1.87793092e-01  5.25848397e-01 -1.25493178e+00  1.97302574e+00]
 [-4.42561890e+02  3.51473337e+01 -2.13766211e+01  1.87919988e+01
  -4.82153941e+00  6.18393353e+00 -8.15131429e+00  4.13130350e+00
   1.82426962e-01  3.81702902e-01 -1.06578501e+00  1.89673915e+00
  -8.87385022e-01  1.52324237e+00  4.40250659e-01  7.52523345e-01
  -1.47740614e+00  1.67340494e+00 -8.67145274e-01  1.23824630e+00]]
[[119.16401089  45.52294802  28.52420539  22.37099628  17.56618646
   17.00234563  15.01655046  13.80996335  13.70452718  10.10141148
    9.16265353   8.14903245   6.97779353   5.88408877   5.75585432
    4.84787398   4.94820668   4.69375034   4.34965304   3.96524215]
 [ 94.38587764  53.28781199  33.07660487  26.07991534  19.66949136
   13.59379339  13.98043676  10.75153258  11.13811352   7.96062535
    7.41648942   7.69679744   5.68304984   5.84380968   5.51277759
    5.73553376   3.93350589   4.76497717   5.1766815    3.75744497]
 [121.97790054  61.1246219   35.74435838  25.00486363  18.08494117
   16.12495676  12.04243182  10.22763624   9.13097552   8.14280756
    7.50139553   6.5485255    6.25000379   5.69170925   5.27612919
    4.59629124   4.17002513   3.97673993   3.70339625   3.83834449]]
```

Hint: You can use the following equations to calculate the model parameter estimates.

$$\widehat{\mu}_{cd} = \frac{\sum_{i=1}^{N} x_{id} 1(y_i = c)}{N_c} \qquad \widehat{\sigma}_{cd} = \sqrt{\frac{\sum_{i=1}^{N} (x_{id} - \widehat{\mu}_{cd})^2 1(y_i = c)}{N_c}}$$

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5. Calculate the score values for the data points in training and test sets using the estimated parameters. (40 points)

```
scores_train = calculate_score_values(X_train, means, deviations, class_priors)
print(scores_train[0:10, :])
```

```
[[ -76.16252841  -77.94721079  -76.00731176]
 [ -78.68514038  -87.57996602  -85.48272895]
 [ -78.2822727   -75.69506373  -73.53605448]
 [ -72.0188793   -76.2330159   -73.91766399]
 [ -98.7057529   -96.83983033 -102.94152458]
 [ -81.70186244  -88.71523861  -79.43014351]
 [ -88.07343391  -76.7277871   -79.24623233]
 [ -73.2092584   -71.76877154  -73.03826832]
 [ -73.0508014   -73.00570881  -71.44594548]
 [ -76.25536804  -72.63116668  -74.4927849 ]]
```

```
scores_test = calculate_score_values(X_test, means, deviations, class_priors)
print(scores_test[0:10, :])
```

```
[[-78.00966505 -73.92102709 -71.36026894]
 [-79.79540559 -84.74033697 -79.87462665]
 [-90.07608202 -80.4430225  -79.45522577]
 [-78.0387075  -68.98702688 -70.05055299]
 [-74.40944488 -76.02196354 -78.97635049]
 [-73.69710979 -77.45126941 -74.90681524]
 [-76.46099798 -77.79114222 -75.60412436]
 [-84.31692172 -79.71213785 -76.26742241]
 [-73.58736818 -82.48264708 -81.17346908]
 [-79.52688944 -78.51314149 -75.33581693]]
```

Hint: You can use the following equation to calculate the score values.

$$
\begin{aligned}
g_c(\boldsymbol{x}) &= \log\left[\prod_{d=1}^{D} \widehat{p}(x_d|y=c)\right] + \log\widehat{\Pr}(y=c) \\
&= \log\left[\prod_{d=1}^{D}\left(\frac{1}{\sqrt{2\pi\widehat{\sigma}_{cd}^2}}\exp\left(-\frac{(x_d-\widehat{\mu}_{cd})^2}{2\widehat{\sigma}_{cd}^2}\right)\right)\right] + \log\widehat{\Pr}(y=c)
\end{aligned}
$$

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6. Calculate the confusion matrices for the data points in training and test sets using the calculated score values. (20 points)

```
[[118   4  19]
 [  7  52  19]
 [  8  11  72]]
Training accuracy is 78.06%.
[[37  1  7]
 [ 3 15  7]
 [ 6  6 18]]
Test accuracy is 70.00%.
```

What to submit: You need to submit your source code in a single file (`.py` file). You are provided with a template file named as `0099999.py`, where `99999` should be replaced with your 5-digit student number. You are allowed to change the template file between the following lines.

```
# your implementation starts below

# your implementation ends above
```

How to submit: Submit the file you edited to LearnHub by following the exact style mentioned. Submissions that do not follow these guidelines will not be graded.

Late submission policy: Late submissions will not be graded.

Cheating policy: Very similar submissions will not be graded.
